

voyAGER: free web interface for the analysis of age-related gene expression alterations in human tissues

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Abstract

We herein introduce voyAGER, an online graphical interface to explore age-related gene expression alterations in 48 human tissues. voyAGER offers a visualization and statistical toolkit for the finding and functional exploration of sex- and tissue-specific transcriptomic changes with age. In its conception, we developed a novel bioinformatics pipeline leveraging RNA sequencing data, from the GTEx project, for more than 700 individuals.

voyAGER reveals transcriptomic signatures of the known asynchronous aging between tissues, allowing the observation of tissue-specific age-periods of major transcriptional changes, that likely reflect so-called digital aging, associated with alterations in different biological pathways, cellular composition, and disease conditions.

voyAGER therefore supports researchers with no expertise in bioinformatics in elaborating, testing and refining their hypotheses on the molecular nature of human aging and its association with pathologies, thereby also assisting in the discovery of novel therapeutic targets. voyAGER is freely available at <https://compbio.imm.medicina.ulisboa.pt/voyAGER>

eLife assessment

This work presents a **valuable** online platform designed to facilitate the exploration of genes and genetic pathways implicated in human aging. Leveraging a new inference methodology, the tool enables the identification and visualization of key genes and tissues impacted by aging, facilitating scientific discovery. However, since the methods have not been fully explained, there is only **incomplete** support for the platform, which could be improved upon for usability and reproducibility.

Introduction

The aging-associated progressive loss of proper tissue homeostasis maintenance makes age a prevalent risk factor for many human pathologies, including cancer, neurodegenerative and

cardiovascular diseases ^{(1)–(3)}. A better comprehension of the molecular mechanisms of human aging is thus required for the development and effective application of therapies targeting its associated morbidities.

Dynamic transcriptional alterations accompany most physiological processes occurring in human tissues ⁽⁴⁾. Transcriptomic analyses of tissue samples can thus provide snapshots of cellular states therein and insights into how their modifications over time impact tissue physiology. A small proportion of transcripts has indeed been shown to vary with age in tissue-⁽⁵⁾ and sex-specific ^{(6)–(11)} manners. Such variations reflect deregulations of gene expression that underlie cellular disfunctions ⁽⁵⁾.

Many studies analyzed the age-related changes in gene expression in rodent tissues ^{(12)–(18)}, emphasizing the role in aging of genes related to inflammatory responses, cell cycle or the electron transport chain. However, while it is possible to monitor the modifications in gene expression over time in those species by sequencing transcriptomes of organs of littermates at different ages, as a close surrogate of longitudinality, such studies cannot be conducted in humans for obvious ethical reasons. Indeed, most studies aimed at profiling aging-related gene expression changes in human tissues focused on a single tissue (e.g. muscle ^{(19)–(21)}, kidney ⁽²²⁾, brain ^{(23)–(27)}, skin ^{(28),(29)}, blood ^{(30),(31)}, liver ⁽³²⁾, retina ⁽³³⁾) and/or limited to a comparison between young and old individuals ^{(20),(21),(28),(32),(33)}, failing to fully capture the changes of tissue-specific gene expression landscape along aging ⁽⁵⁾. A few studies were nonetheless led on more than one tissue in humans, from post-mortem samples ^{(34),(35)} and biopsies ^{(36),(37)}, and in mice ^{(18),(36)} and rats ⁽³⁸⁾. The age-related transcriptional profiles derived therein, either from regression ^{(18),(34),(35),(37)} or comparison between age groups ^{(36),(38)}, highlight an asynchronous aging of tissues (discussed in ⁽³⁹⁾), with some of them more affected by age-related gene expression changes associated with biological mechanisms known to be impacted by aging such as mitochondrial activity or metabolic homeostasis. In particular, tissue-specific periods of major transcriptional changes in the fifth and eighth decades of human lifespan have been revealed ⁽³⁴⁾, reflecting the so-called digital aging ⁽³⁹⁾ and consistently with what is observed in mice ^{(17),(18)}. Furthermore, despite outlining the tissue specificity of the transcriptomic signatures of human aging, some synchronization was found between tissues like lung, heart and whole blood, which exhibit a co-aging pattern ⁽³⁵⁾. Nevertheless, as each study followed its own specific procedures, from sample collection to data processing, results from their analyses are hardly comparable across studies.

Processed data from those studies have not been made easily accessible and interpretable to researchers lacking computational proficiency but aiming to use them to test their novel hypotheses. To fill this void, we have developed voyAGER, a web application providing flexible visualization of comprehensive functional analyses of gene expression alterations occurring in 48 human tissues with age in each sex. We leverage the large RNA-seq dataset from the Genotype-Tissue Expression (GTEx) project ⁽⁴⁰⁾, encompassing post-mortem tissue samples from hundreds of donors aged from 20 to 70 years, with a pipeline for gene expression profiling with an optimized temporal resolution. voyAGER allows to investigate aging from two perspectives: (i) gene-centric – how each gene's tissue-specific expression progresses with age; (ii) tissue-centric – how tissue-specific transcriptomes change with age. Additionally, voyAGER enables the examination of modules of co-expressed genes altered with age in 4 tissues (brain, skeletal muscle, heart (left ventricle), whole blood), namely their enrichment in specific cell types, biological pathways, and association with diseases. We therefore expect voyAGER to become a valuable support tool for researchers aiming to uncover the molecular mechanisms underlying human aging. Moreover, being open-source, voyAGER can be adapted by fellow developers to be used with alternative datasets (e.g. from other species) or to incorporate other specific functionalities. voyAGER is freely available at <https://compbio.imm.medicina.ulisboa.pt/voyAGER>

Results

voyAGER's interactive exploration of tissue-specific gene expression landscapes in aging is based on the sequential fitting of linear models (v. Materials and Methods) to estimate, for each gene in each tissue (Figure 1):

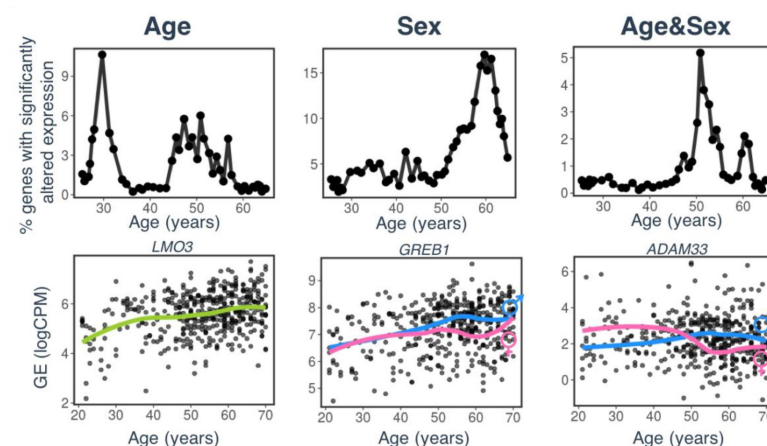


Figure 1:

voyAGER profiles tissue-specific age-associated changes in gene expression and their differences between sexes.

For each of the 48 human tissues in GTEx, gene expression was linearly modelled in windows spanning 16 years centered in consecutive years of age, to estimate the effects thereon of *Age*, *Sex* and the interaction between them (*Age&Sex*), i.e., how the *Sex* effect changes with age, equivalent to how the *Age* effect differs between sexes (v. Materials and Methods). In each window (i.e., for each age in consecutive years), the percentage of genes with expression significantly altered by each of those effects gives their respective transcriptomic impact (upper panels). voyAGER thereby identifies the age periods at which major gene expression changes occur in each tissue. For example, in subcutaneous adipose tissue: major age-related transcriptional alterations are found at 30 and around 50 years of age (upper left panel), illustrated by the behavior of *LMO3* (bottom left panel); major gene expression differences between males and females happen at around 60 years of age (upper center panel), as illustrated by *GREB1* (bottom center panel); major differences between sexes in age-related gene expression alterations happen at around 50 years of age (upper right panel), as illustrated by *ADAM33* (bottom right panel). Solid loess lines in the bottom panels (green for all donors, pink for females, blue for males). Gene expression (GE) in log₂ of counts per million (logCPM).

1. the *Age* effect, i.e., how the age-associated changes in gene expression evolve with age itself;
2. the *Sex* effect, i.e., how the differences in gene expression between sexes evolve with age;
3. the *Age&Sex* interaction effect, i.e., how the *Sex* effect changes with age, equivalent to how the *Age* effect differs between sexes; in other words, how the age-associated

changes in differences in gene expression between sexes (or how the differences between sexes of age-associated changes in gene expression) evolve with age.

We named our approach Shifting Age Range Pipeline for Linear Modelling (ShARP-LM). Briefly, this method carries out differential gene expression analyses (with gene expression as a function of the donors' Age, Sex and Age&Sex interaction) in age windows spanning 16 years centered in consecutive years of age. By considering the percentage of genes differentially expressed in each age range, we highlight age periods of major tissue-specific transcriptomic alterations (Figure 1).

Gene-centric analyses of human tissue-specific expression changes across age

The progression of tissue-specific expression of a specific gene across age can be examined in votAGER's Gene tab. By entering its symbol in the Gene selector, the user has access to graphical summaries of the gene's tissue-specific expression (sub-tab Profile) (Figure 2A) and of the significance age-related changes in its expression due to the Age, Sex, and Age&Sex (sub-tab Alteration) (Figure 2B) across age. Results can be displayed in a heatmap for all tissues or in a scatter plot for a chosen individual tissue. When the gene is studied in a single tissue, the user can graphically and statistically profile the association of the donors' sex and reported conditions (e.g., history of heart attack or pneumonia) with the gene's expression profile (Figure 2C). A table summarizing the donors' metadata is also shown (Figure 2C). The user can interactively select donors of interest on the scatter plot and further examine their information in the automatically subsetting table.

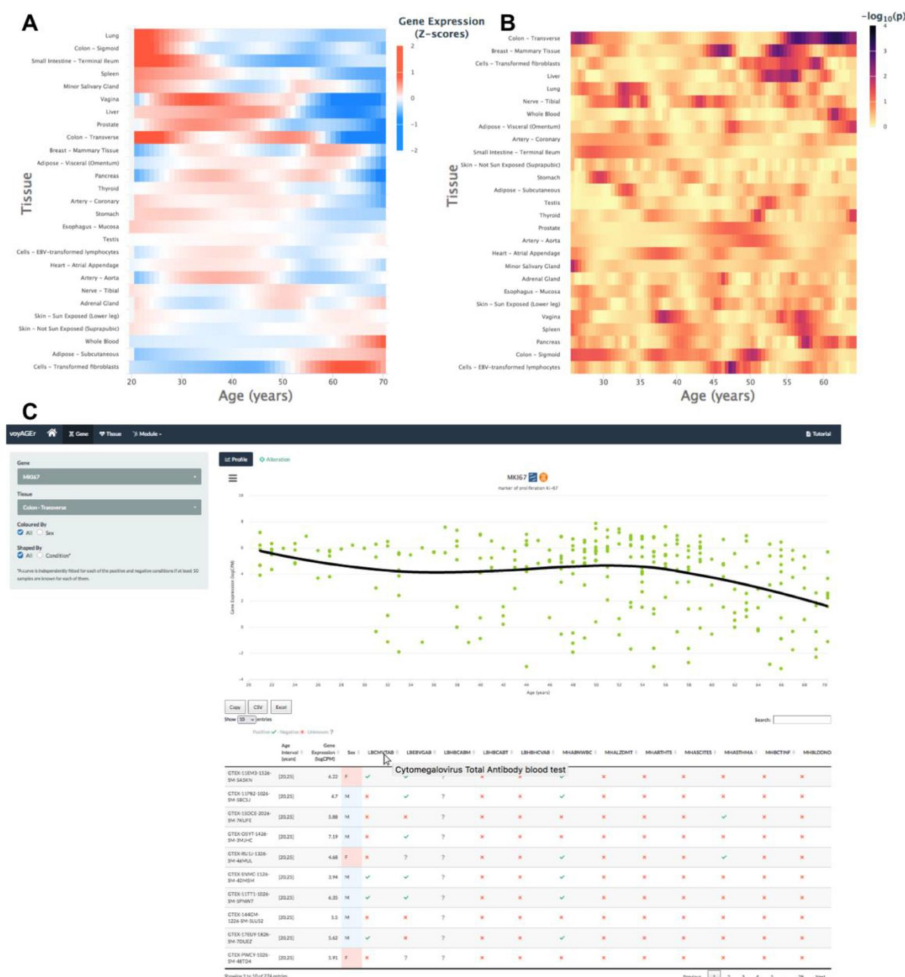


Figure 2:

Gene-centric analyses of expression alterations across age

A. Heatmap of MKI67 expression across tissues over age.

B. Heatmap of significance of Age-associated MKI67 expression alterations over age. p-values are for the moderated t-statistics of differential gene expression associated with the Age effect (v. the ShARP-LM approach in Materials and Methods). Notably, major transcriptional changes are observed in transverse colon (mid 20's and after 55), breast (mid/late 40's and early/mid 50's), and liver (mid 50's).

C. votAGER's Gene tab interface. MKI67 expression in transverse colon is inspected. Donors' information is shown in a table and the scatter plot can be interac-

Cellular senescence, a stress-induced cell cycle arrest limiting proliferation of potentially oncogenic cells but progressively creating an inflammatory environment in tissues as they age, is an example of a process whose molecular mechanisms are of particular interest to aging researchers ^{(41),(42)}. *CDKN2A*, encoding cell cycle regulatory protein p16^{INK4A} that accumulates in senescent cells ^{(43),(44)}, has its expression increased with age in the vast majority of tissues profiled (**Supplementary Figure 1A**). Similarly, reduced levels of proliferation markers, such as *PCNA* ⁽⁴⁵⁾ and *MKI67* ⁽⁴⁶⁾, can be studied as putative markers of aging of certain tissues. These genes have their expression significantly altered with age in transverse colon and display a similar expression profile (decreasing from 20 to 30 years old (y.o.), constant until 55 y.o. and decreasing in older ages) (**Figure 2C** and **Supplementary Figure 1B**). However, these trends appear to vary according to the donor's history of pneumonia (**Supplementary Figure 1B**), illustrating voyAGER's use in helping to find associations between gene expression and age-related diseases.

On a different note, sex biases have been reported in for the expression of *SALL1* and *KAL1* in adipose tissue and lung, respectively ⁽¹⁰⁾. voyAGER allows to not only recapitulate those observations but also estimate that these differences occur mostly in older ages (**Supplementary Figure 2**).

Tissue-specific assessment of gene expression changes across age

Peaks

The landscape of global tissue-specific gene expression alteration across age can be examined in voyAGER's *Tissue* main tab. A heatmap displaying, for all tissues, the statistical significance over age (v. Materials and Methods) of the proportion of genes altered with *Age*, *Sex* or *Age&Sex* interaction (depending on the user's interest) is initially shown (**Figure 3A**), illustrating the aforementioned asynchronous aging of tissues observed for humans and rodents ^{(18),(32)–(36),(39)}.

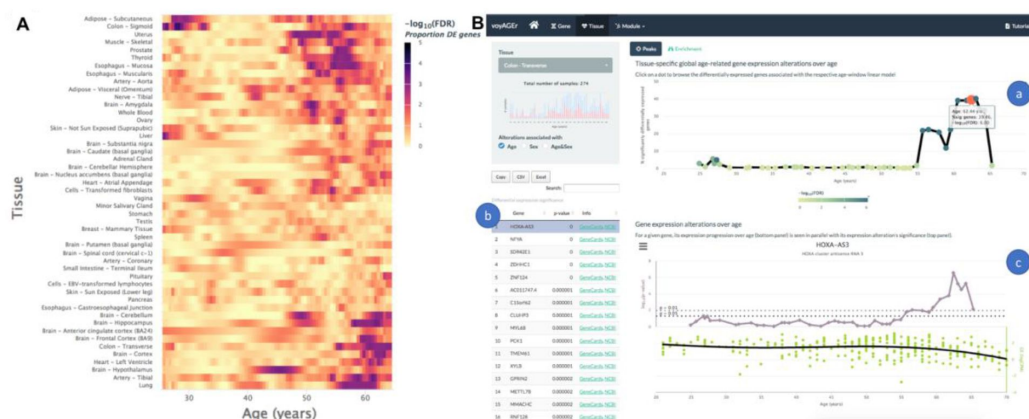


Figure 3:

Tissue-specific assessment of gene expression changes across age

A. Heatmap of significance (FDR based on random permutations of age, v. Materials and Methods) over age of the proportion of genes with expression significantly altered with *Age* in the 48 analyzed tissues.

B. Exploration of gene expression changes across age in transverse colon.

- a. Percentage of genes with significantly altered expression with *Age* over age. Three peaks at 27, 55 and 62 years old are noteworthy.
- b. Clicking on the dot of a specific age (62.44 y.o. in plot *a*) gives access to the list of the most differentially expressed genes at that age, ordered by statistical significance of expression changes (p-value of moderated t-statistic).
- c. Plot of expression of the chosen top gene in the table in *b* across age (bottom) in parallel with the significance of its expression alterations with *Age*. The expression of *HOXA-AS3* significantly decreases at around 62 years old, concomitantly with the third peak of transcriptomic changes with *Age*.

The user can then spot the age periods with the most significant gene expression alterations in a selected tissue (**Figure 3B a**), the so-called *Peaks* naming the sub-tab, and identify the associated differentially expressed genes (**Figure 3B b**). The user can also plot the expression of a given gene of interest across age together with the significance of its expression modification with *Age*, *Sex* or *Age&Sex* (**Figure 3B c**).

As an example, the transverse colon appears to go through three main periods of transcriptional changes: one minor at around 27 y.o. (~5 % of genes differentially expressed (DEG)) and two major at around 55 y.o. (~21 % DEG) and 62 y.o. (~40 % DEG) (**Figure 3B a**). Interestingly, some of the genes most differentially expressed in the second peak at ~55 y.o. appear to have their expression modified only at this precise age period (e.g. *E2F1*, *KIF24*, *TRMU*, *CRIM1*). Similarly, genes such as *HOXA-AS3*, *NFYA*, *ZDHHC1*, and *MYL6B* (**Figure 3B b**), are specifically differentially expressed in the third peak at ~62 y.o. (**Figure 3B c**). This particularity is also found for the minor first peak. It therefore seems that different sets of genes drive the periods of major transcriptional changes, which begs assessing if they reflect the activation of distinct biological processes.

Enrichment

The user can thus explore the biological functions of the set of genes underlying each peak of transcriptomic changes by assessing their enrichment in manually curated pathways from the Reactome database ⁽⁴⁷⁾. voyAGER performs Gene Set Enrichment Analysis (GSEA) ⁽⁴⁸⁾ and the user can visualize heatmaps displaying the evolution over age of the resulting normalized enrichment score (NES, reflecting the degree to which a pathway is over- or under-represented in a subset of genes) for the given tissue, effect (*Age*, *Sex* or *Age&Sex*) and all or some user-specified Reactome pathways (**Figure 4A**). To reduce redundancy and facilitate the understanding of their biological relevance, we clustered those pathways into families that also include Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways ⁽⁴⁹⁾ and Gene Ontology (GO) Biological Processes of level 3 ⁽⁵⁰⁾. Briefly, we clustered gene sets from the 3 sources based on the overlap of their genes (v. Materials and Methods), thereby creating families of highly functionally related pathways. Taking advantage of the complementary and distinct terminology in Reactome, KEGG and GO, the user can interpret each family's broad biological function by looking at the word cloud of its most prevalent terms, still being able to browse the list of its associated pathways (**Figure 4A**). The peaks of transcriptomic changes can also be examined for enrichment in a user-provided gene set (**Figure 4B**). To illustrate this procedure, we inspected the aforementioned three aging peaks of the transverse colon for significant enrichment in genes retrieved from Senesquest ⁽⁴²⁾ whose link with senescence is supported by at least 4 sources, having observed it in the first (~27 y.o.) and second (~55 y.o.) peaks (**Figure 4B**).



Figure 4:

Tissue-specific assessment of pathway expression changes across age

A. Heatmap of enrichment (above), given by GSEA's normalized enrichment score (NES), of Reactome pathways in genes differentially expressed with *Age* over age. Pathways are classified into broad families (upper bar), whose associated biological processes and functions can be learnt from the word clouds of their most frequent terms (below). Only Reactome pathways found significantly associated ($FDR \leq 0.01$) with gene expression alterations in at least one age window are included.

B. Enrichment, given by the significance of Fisher's tests, of a user-provided gene set in genes differentially expressed (based on a user-defined p-value threshold) with *Age* over age. Here, the given gene set is composed of genes from Senequest ⁽⁴²⁾ whose link with senescence is supported by at least 4 sources. The first and second peaks of *Age*-associated transcriptomic changes in the transverse colon appear to be enriched in senescence-related genes.

Modules of co-expressed genes

voyAGER also allows functional analyses of modules of co-expressed genes (i.e. genes with highly correlated expression across samples, defined by weighted correlation network analysis ⁽⁵¹⁾) altered with age. Genes in the same module are likely to be coregulated and share biological functions or associations with phenotypical or pathological traits ⁽⁵²⁾. Those modules may also act as markers of core transcriptional features of cellular activity and identity ⁽⁵³⁾.

Concretely, voyAGER enables the user to visualize how the expression of modules of genes that are associated with a specific cell type or disease progresses over age in a specific tissue. In the example of **Figure 5B**, the “midnight blue” module, gathering 11 genes co-expressed in the heart's left ventricle, shows significant enrichment in fibroblast markers, namely collagens. This module's expression in the left ventricle is steady until ~55 years of age and appears to increase later in life, perhaps reflecting the known age-related changes of the collagen network of the human myocardium ⁽⁵⁴⁾.

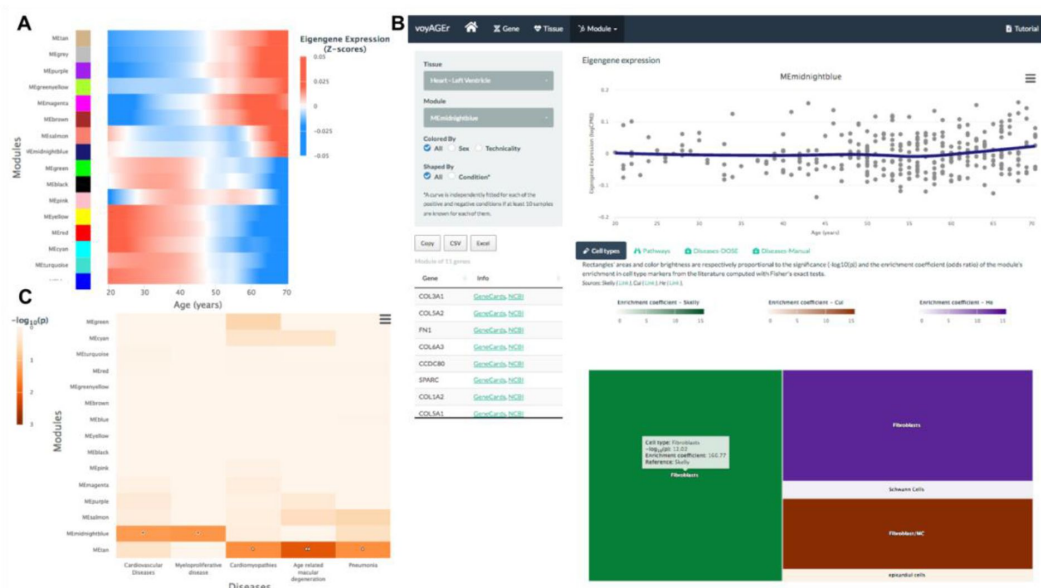


Figure 5:

Tissue-specific assessment of age-associated progression of modules of co-expressed genes

A. Heatmap of eigengene expression for all the inferred modules of co-expressed genes in the heart's left ventricle over age.

B. Scatter plot (above) of eigengene expression over age, in all left ventricle samples, for a selected module of 11 genes. The eigengene expression is derived from the 1st component of the single value decomposition of its genes' expression. This module was analysed, with Fisher's tests, for enrichment in cell types, based on markers from the literature (Skelly⁽⁸⁴⁾, Cui⁽⁸³⁾ and He⁽⁸²⁾, surnames of the first authors of the studies the markers are retrieved from) and found to be associated with fibroblasts, as revealed by the TreeMap below (where each rectangle's area and darkness are proportional to the significance of its association with a cell type and its color linked to the markers' source study).

C. Heatmap of association of the modules with five selected diseases, computed with the DOSE package⁽⁵⁷⁾.

After selecting one tissue of interest, the user has, for each module, access to four levels of information: (i) *Expression*: its eigengene expression progression over age; (ii) *Cell types*: its enrichment in specific cell types, based on marker genes found literature, (iii) *Pathways*: its enrichment in Reactome pathways; (iv) *Diseases*: its enrichment in disease markers, based on gene-disease associations from DisGeNET^{(55),(56)}, calculated with both the DOSE package⁽⁵⁷⁾ and with Fisher's tests (Figure 5C).

By default, for each tissue, results are displayed in the form of heatmaps showing how all modules evolve with age (Figure 5A). The user has also the possibility to select a module of interest and see its eigengene progression over age in a scatter plot (Figure 5B), lists of its association with diseases and pathways ordered by significance, and a TreeMap for its cell type enrichment (Figure 5B). Like in the *Gene* tab, the user can separate donors based on their sex and previous condition in the scatter plot of the eigengene expression progression.

When a significant enrichment results from a Fisher's test, the associated contingency table is shown for its clearer interpretation.

Discussion

voyAGER provides a framework to examine the progression over age of gene expression in a broad range of human tissues, thereby being a novel valuable resource for the aging research community. In particular, it helps to identify tissue-specific age periods of major transcriptomic alterations. The results of our analyses show the complexity of human biological aging by stressing its tissue specificity⁽³⁴⁾ and non-linear transcriptional progression throughout lifetime, consistent with previous results from both proteomic⁽⁵⁸⁾ and transcriptomic^{(28),(34)} analyses. By revealing and annotating the age-specific transcriptional trends in each tissue, voyAGER aims at assisting researchers in deciphering the cellular and molecular mechanisms associated with the age-related physiological decline across the human body.

Besides, voyAGER, to our knowledge unprecedentedly, scrutinizes and visually displays the tissue-specific differences in gene expression between biological sexes across age. Biological sex is an important variable in the prevalence of aging-associated diseases, as well as in their age of onset and progression^{(6),(7),(59)} and tissue-specific sex-related biases in gene expression had been reported^{(8)–(11)}. By profiling the age distribution of those biases, voyAGER will thus help to better understand their influence in the etiology of the sex specificities of human aging. For instance, we were able to corroborate findings on the sex-differential transcriptome of human adults by Gershoni et al.⁽⁹⁾, with voyAGER emphasizing its tissue-specificity and allowing to discriminate the ages at which sex-related biases appear to be more prevalent ([Supplementary Figure 3](#)).

One of the limitations of voyAGER is that most GTEx tissue donors had health conditions and their frequency increased with age, preventing us from defining a class of healthy individuals and identifying age-associated transcriptomic changes that could be more confidently proposed to happen independently of any disease progression. On one hand, we expect the large sample sizes and the biological variability between individuals, reflected on the diversity of combinations of conditions, to mitigate major confounding effects. On the other hand, voyAGER allows users to assess how tissue-specific gene expression trends vary according to the donors' history of the different conditions ([Supplementary Figure 1B](#)), therefore helping to find associations between gene expression and age-related diseases.

The development of voyAGER was accompanied by that of a pipeline, ShARP-LM, that facilitates the holistic depiction of the transcriptional landscapes of adult human tissues upon aging with a yearly age resolution. We take advantage of the comprehensiveness of the collection of transcriptomes from post-mortem human tissues from the GTEx project to make our analyses a valid surrogate of a currently undoable longitudinal study. It confers our method enough statistical robustness to mitigate the inter-individual, overcome the imbalance in the numbers of male and female samples and deal with the non-uniform distribution of the donors' ages. Indeed, we found no confounding between the distribution of samples' ages and the trend of gene expression progression over age in any tissue.

However, the distribution of donor's ages does affect the statistical power of detection of transcriptional alterations and we are therefore more likely to find significant ones in age ranges that are more abundant in samples ([Supplementary Figure 4](#)). This limitation is to be overcome by the accumulation of transcriptomes of human tissues in public databases, promising a gradually increase in accuracy and age resolution with which human transcriptomic aging can be profiled. Similarly, the accumulation of single-cell

transcriptomes in public databases is yielding better gene markers for a growing diversity of human cell types, increasing the usefulness of exploiting bulk transcriptomes in studying the impact of aging on the cellular composition of human tissues, for which the co-expression module approach in voyAGER provides a proof-of-concept.

As voyAGER is open source, it is envisaged as an ever-evolving resource, that can be adapted by researchers to accommodate the aforementioned new data and expanded to provide new functionalities (e.g. integration with perturbagen data for the inference of molecular causes of observed gene expression alterations or small molecules for therapeutic purposes ^{(60),(61)}).

As an *in silico* approach with no experimental validation for its results, voyAGER is meant to be a discovery tool, supporting biologists of aging in the exploration of a large transcriptomic dataset, thereby generating, refining or preliminary testing hypotheses that will pave the way for subsequent experimental work. It can be an entry point for projects aiming at better understanding the tissue- and sex-specific transcriptional alterations underlying human aging, to be followed by more targeted studies focusing on the functional roles of the most promising markers identified therein in the physiology of aging. Those marker genes can contribute to the development of more robust and cross-tissue gene signatures of aging ⁽⁶²⁾ and the expansion of age-related gene databases ^{(57),(63)}.

Moreover, the observed diverse and asynchronous changes in gene expression between tissues over the human adult life provides potentially relevant information for the design of accurate diagnostic tools and personalized therapies. On one hand, finding those changes' association with specific disorders could have a prognostic value by enabling the identification of their onset before their clinical symptoms' manifestation ⁽³⁶⁾. On the other hand, computational screening of databases of genetic and pharmacologically induced human transcriptomic changes could help to infer their molecular causes and shed light on candidate drugs to delay their effects ^{(60),(61),(64),(65)}.

Methods

Development's platform

Data analysis was performed in R (version 3.6.3) and the application developed with R package Shiny⁽⁶⁶⁾. voyAGER's outputs are plots and tables, generated with R packages highcharter⁽⁶⁷⁾ and DT⁽⁶⁸⁾, respectively, that can easily be downloaded in standard formats (png, jpeg, and pdf for the plots; xls and csv for the tables).

voyAGER was deployed using Docker Compose and ShinyProxy 2.6.0 in a Linux virtual machine (64GB RAM, 16 CPU threads and 200GB SSD) running in our institutional computing cluster. The source code for voyAGER is available at GitHub (<https://github.com/nunolbmorais/voyAGER>).

Read count data pre-processing

We downloaded the matrix with the RNA-seq read counts for each gene in each GTEx v7 sample from the project's data portal (<https://www.gtexportal.org/>) ⁽⁴⁰⁾. We then pre-processed the read count data for each tissue separately. We started by filtering out the uninformative genes with very low expression across samples. For that purpose, we normalised read counts for library size, to account for the differences in sequencing depth between samples, by transforming them into counts per million (CPM) with function `cpm` from R package edgeR⁽⁶⁹⁾. Only genes with at least 1 CPM in at least 40% of the samples were

kept for analysis (the number of genes analysed for each tissue can be found in [Supplementary Table 1](#)). Read counts for those kept genes were used to calculate normalization factors to scale the raw library sizes, using function `calcNormFactors` from `edgeR` that implements the trimmed mean of M-values ⁽⁷⁰⁾, and normalised accordingly and subsequently log-transformed with the `voom` function ⁽⁷¹⁾ from package `limma` ⁽⁷²⁾, which also estimates mean-variance relationships to be used in linear modelling (v. ShARP-LM below). Normalized gene expression was thereby ready to be analyzed and featured in the plots in log2 of CPM (logCPM), this log-transformation yielding more manageable nearly normal distributions.

Finally, from the 53 tissues available from GTEx v7, we discarded five (renal cortex, fallopian tube, bladder, ectocervix, endocervix, chronic myeloid leukemia cell line) due to a low (<50) number of respective samples.

ShARP-LM

To model the changes in gene expression with age, we developed the Shifting Age Range Pipeline for Linear Modelling (ShARP-LM). For each tissue, we fitted linear models to the gene expression of samples from donors with ages within windows with a range of 16 years shifted through consecutive years of age (i.e. in a sliding window with window size = 16 and step size = 1 years of age). As samples at the ends of the dataset's age range would be thereby involved in fewer linear models, we made the window size gradually increase from 11 to 16 years when starting from the “youngest” samples and decrease from 16 to 11 years when reaching the “oldest” ([Supplementary Figure 5](#)).

Function `lm` from `limma` was used to fit the following linear model for gene expression (GE):

$$GE \sim \alpha.Age + \beta.Sex + \chi.Age\&Sex + \varepsilon$$

For each gene, α , β and χ are the coefficients to be estimated for the respective hypothesized effects and ε is the residual. For each sample, *Age* in years and *Sex* in binary were centered and *Age&Sex* interaction given by their product.

For each gene in each model (i.e., each age window in each tissue), we retrieved the t-statistics of differential expression associated with the three variables and their respective p-values. We considered the average age of the samples' donors within the age window as the representative age of the observed expression changes.

In summary, for a given tissue and variable (*Age*, *Sex* and *Age&Sex*), ShARP-LM yields t-statistics and p-values over age for all genes, reflecting the magnitude and significance of the changes in their expression throughout adult life.

Gene-centric visualization of tissue-specific expression changes across age

For visualization purposes, the trend of each gene's expression progression over age in each tissue was derived through Local Polynomial Regression Fitting, using R function `loess` on logCPM values ([Figures 1, 2C, 3B](#)). For summarizing in a heatmap a given gene's expression across age in multiple tissues ([Figure 2A](#)) or the expression of several genes across age in a given tissue, each gene's regression values are centered, but not scaled, with R function `scale`.

For summarizing in a heatmap the significance of a given gene's expression changes over age in multiple tissues (**Figure 2B**), cubic smoothing splines are fitted to $-\log_{10}(p)$, p being the t-statistic's p-value, with R function `smooth.spline`.

Tissue-specific quantification of global transcriptomic alterations across age

To assess the global transcriptomic impact of each of the three modelled effects in each tissue across age, we then analyzed the progression over age (i.e. over consecutive linear models) of the percentage of genes whose expression is significantly altered (t-statistic's p-value ≤ 0.01) by each effect (**Figure 3B**). To evaluate the significance of each percentage and see if high ones can be confidently associated with major transcriptomic alterations, we controlled for their false discovery rate (FDR) by randomly permuting the samples' ages within each age window one million times and performing ShARP-LM on each randomised dataset. We were then able to associate an FDR to each percentage of differentially expressed genes by comparing it with the distribution of those randomly generated (**Figure 3B**).

Gene Set Enrichment Analysis (GSEA)

For each *Peak* of significant gene expression modifications, we performed GSEA⁽⁴⁸⁾ on the ordered (from the most positive to the most negative) t-statistics of differential expression for the respective tissue and age, using R package `fgsea`⁽⁷³⁾ and the Reactome database⁽⁴⁷⁾. The resulting normalized enrichment score (NES) for each pathway was used in subsequent analyses as a metric of its over- or down-representation in the *Peak*.

Families of pathways

To reduce pathways' redundancy and facilitate the assessment of their biological relevance in results' interpretation, we created an unifying representation of pathways from Reactome⁽⁴⁷⁾ and KEGG⁽⁴⁹⁾, and level 3 Biological Processes from GO⁽⁵⁰⁾, by adapting a published pathway clustering approach⁽⁷⁴⁾ to integrate them into families.

The approach relies on the definition of a hierarchy of pathways based on the numbers of genes they have in common. For each two pathways P_i and P_j , respectively containing sets of genes G_i and G_j , we computed their overlap index⁽⁷⁴⁾, defined as following:

$$O_{ij} = |G_i \cap G_j| / \min(|G_i|, |G_j|)$$

Where $|G_i|$ is the number of genes in set G_i and $|G_i \cap G_j|$ is the number of genes in common between G_i and G_j . $O_{ij} = 1$ would therefore indicate that P_i and P_j are identical in gene composition or that one is a subset of the other. On the contrary, $O_{ij} = 0$ would mean that P_i and P_j have no genes in common. To ease the computational work, we removed from the analysis pathways that are subsets of larger pathways (i.e. each pathway whose genes are all present in another pathway).

From the O_{ij} matrix, from which each row is a vector with the gene overlaps of pathway I with each of all the pathways, we computed the matrix of Pearson's correlation between all pathways' overlap indexes with R function `cor`. That matrix was finally transformed into Euclidean distances with R function `dist`, allowing for pathways to be subsequently clustered with the complete linkage method with R function `hclust`. The final dendrogram was

empirically cut in 10 clusters ([Supplementary Figure 6](#)). One of them, not including Reactome pathways, was dismissed from the analysis. Pathways that initially excluded from the computation for being subsets of others were finally added to the clusters of their respective *parent* pathways. Each *daughter* pathway with more than one parent was assigned to the cluster of the parent with the smallest number of genes, thereby maximizing the daughter-parent similarity. The data.table R package, for fast handling of large matrices, was used in this analysis ⁽⁷⁵⁾.

Gene co-expression modules

Gene co-expression modules were defined with R package WGCNA⁽⁵¹⁾. For each tissue, genes were first filtered based on the variance of their expression across samples (*variable A*), with lowly variant ones being discarded from subsequent co-expression analyses. Then, an adjacency matrix was computed with the adjacency function. To this end, we first computed the matrix of Pearson's correlation between all selected genes' expressions and then raised all of its values to a power β (soft thresholding) to emphasise stronger correlations. The soft threshold was determined with the pickSoftThreshold function. The thereby generated matrix ($| \text{correlation} |^\beta$) was then converted into a dissimilarity matrix based on the topological overlap measure, determined with function TOMsimilarity. Finally, gene co-expression modules were identified with a dynamic tree-cutting algorithm, implemented with the cutreeDynamic function, that adjusts the minimal number of genes per module (*variable B*) and the height at which the tree is cut (*variable C*).

The definition of those modules of co-expressed genes is highly dependent on the three variables *A*, *B*, and *C*. Thus, we comprehensively tested combinations of them and defined as optimal those that maximized the modules' enrichment in known markers for specific cell types, an approach similar to that reported to have been successfully applied in the revelation of the core transcriptional features of cell types in the human central nervous system ⁽⁵³⁾. In particular, we used human and murine cell type gene signatures, derived from single cell transcriptomic analyses, for brain (cortex) ^{(76)–(79)}, skeletal muscle ^{(80)–(82)}, left cardiac ventricle ^{(82)–(84)}, and whole blood ^{(82),(85),(86)}. Finally, the 40%, 30%, 35%, and 40% most variant genes, the tree cut at a height of 0.96, and a minimum number of 20, 18, 11, and 16, were respectively chosen for the three variables in those four tissues.

Each module is then represented by a set of genes and an eigengene, the first principal component obtained by singular-value decomposition of the expression of the module's genes and representative of its own expression profile.

The enrichment of modules in cell types, Reactome pathways and diseases from DisGeNET ^{(55),(56)} was quantified with Fisher's tests. Disease enrichment was also calculated using the DOSE package ⁽⁵⁷⁾.

Data availability

Processed GTEx v7 RNA-seq data (read count tables) were downloaded from the project's data portal (<https://www.gtexportal.org/>). Donor metadata were obtained from dbGaP - database of Genotypes and Phenotypes (Accession phs000424.v7.p2; project ID 13661). voyAGER's output tables be directly downloaded in standard xls and csv formats.

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Authors' contributions

A.S. and N.L.B.-M. designed the study; A.S. performed all the development and bioinformatics work, under the supervision of N.L.B.-M.; N.S.-A. set up the web and database implementation; A.S. and N.L.B.-M. wrote the manuscript, with contributions from N.S.-A..

Conflict of Interest statement

The authors declare no conflict of interest.

Supplementary Figures and Tables

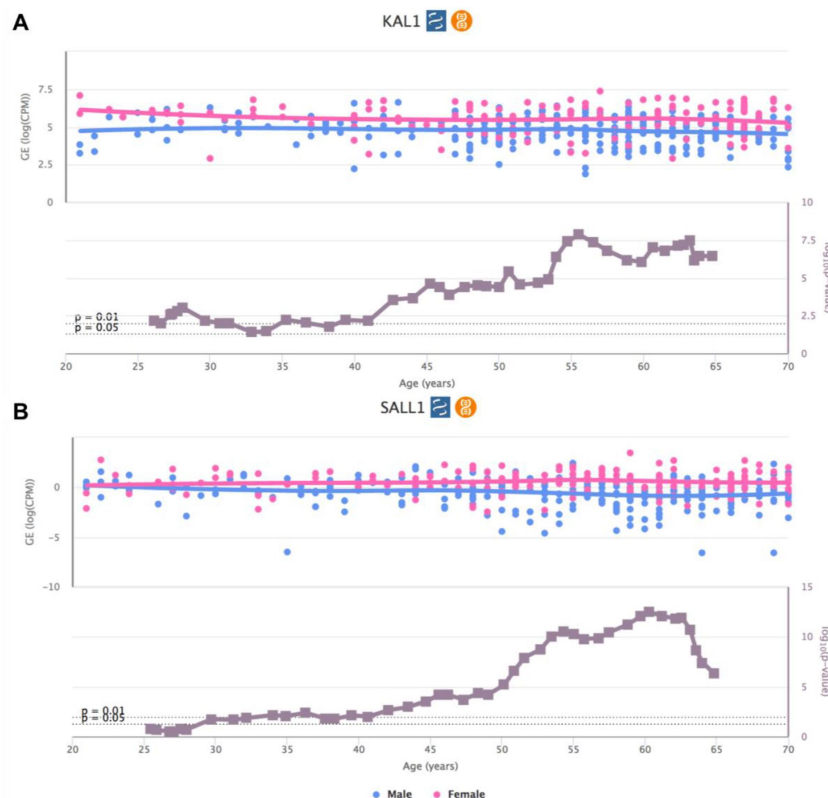


Supplementary Figure 1:

voyAGER's Gene tab interface.

A. Heatmap of *CDKN2A* expression across tissues over age.

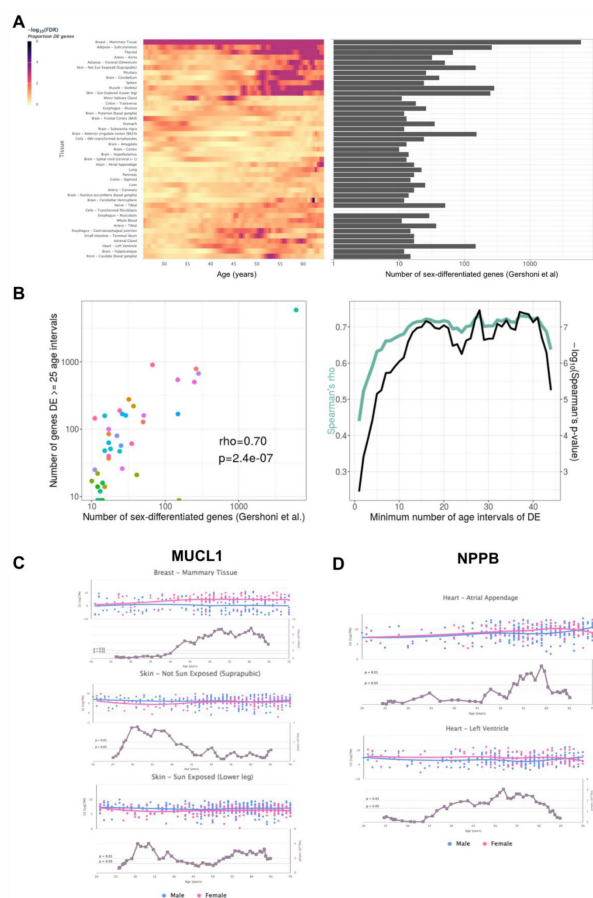
B. *PCNA* expression in transverse colon over age and its association with the donor's history of pneumonia.



Supplementary Figure 2:

Sex-specific *SALL1* (A) and *KAL1* (B) expression alterations over age.

Scatter plots of *SALL1* (A) and *KAL1* (B) expression over age (above), where dots are colored based on the donor's sex (male in blue and female in pink), are shown together with those of the significance of their difference between sexes (below).



Supplementary Figure 3:

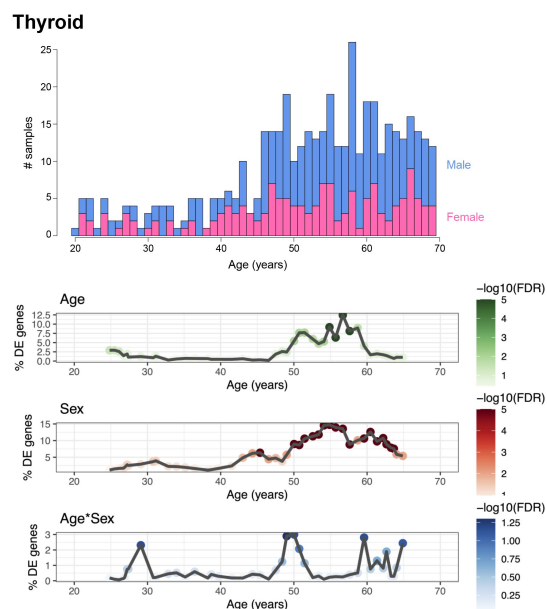
Tissue-specific sex-differentiated expression.

A. Heatmap of significance over age, for each tissue, of the proportion of differentially expressed genes between sexes (left), in parallel with the number of sex-differentiated genes highlighted by Gershoni et al. (v. Figure 1 in ⁽⁹⁾) (right). Mammary is corroborated as the tissue with the most sex-differentiated gene expression. For tissues like adipose, skeletal muscle, skin, and heart, sex-related biases in expression mainly arise in late life.

B. Scatter plot relating, for each tissue (each dot), the number of sex-differentiated genes found by Gershoni et al. with the number of genes found by voyAGER to be differentially expressed between sexes in at least 25 age intervals (left). This correlation is significant regardless of the considered minimum number of age intervals a gene must be differentially expressed in (right). Half visible dots in the left scatter plot represent tissues found to have no sex-differentiated genes by voyAGER (the log scale would give them a minus infinite value on the Y-axis).

C. and D. Two examples of genes, *MUCL1* (C) and *NPPB* (D), whose expression is known to be sex-differentiated in a tissue-specific manner (Figures S7 and 3 in ⁽⁹⁾, respectively). Plots of

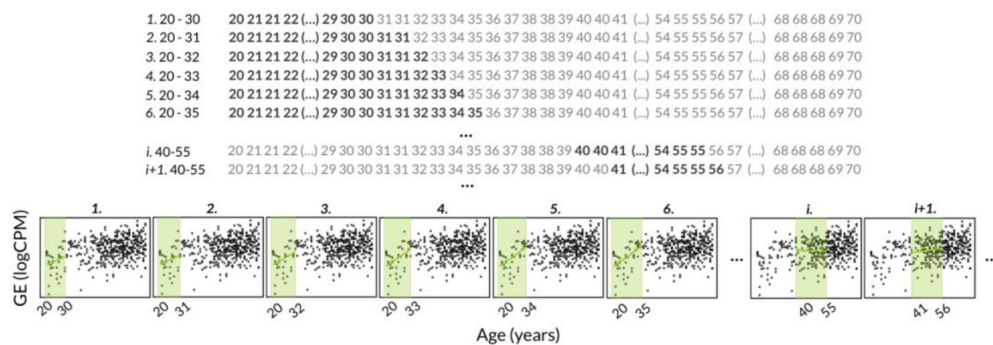
their sex-specific expression across age in parallel with the significance of their differences in expression alterations between sexes are shown for tissues reported in ⁽⁹⁾ as having sex-differentiated gene expression: breast and skin for *MUCL1* and heart for *NPPB*. voyAGER helps to refine the definition of the age periods at which those biases occur: late-life in breast and early-life in skin for *MUCL1*, and late-life in heart for *NPPB*.



Supplementary Figure 4:

Distributions of GTEx donors' ages and its impact on the statistical power of detection of differential gene expression.

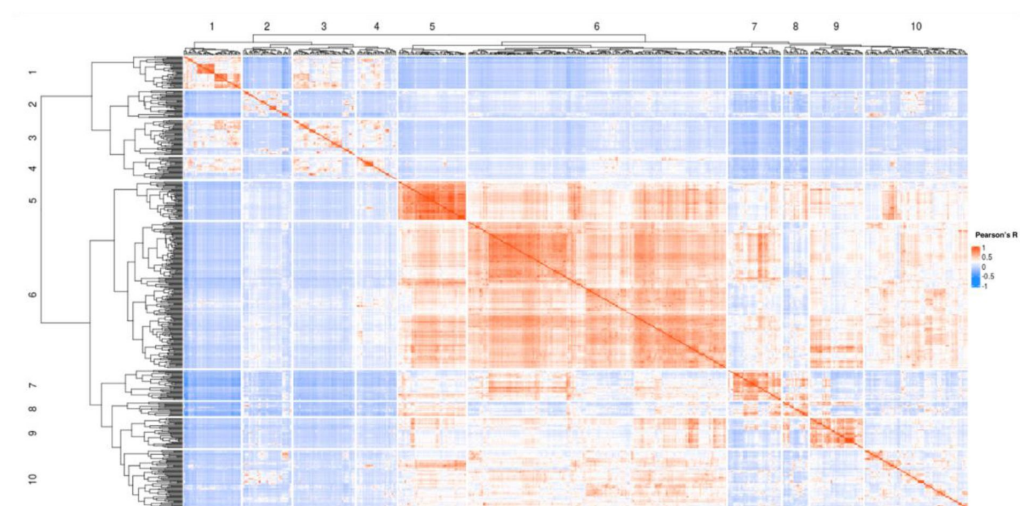
For thyroid as an example tissue, the sex-differentiated distribution of GTEx samples' ages (v7) (above) are shown together with the percentage of differentially expression genes (% DE genes) over age (below) for the three variables (*Age*, *Sex* and *Age&Sex*), as well as their statistical significance, given by the ShARP-LM approach.



Supplementary Figure 5:

Principle of the ShARP-LM method.

ShARP-LM consists of fitting, for each tissue, linear models to the gene expression of samples from donors with ages within windows with a range of 16 years shifted through consecutive years of age, in a sliding window with window size = 16 and step size = 1 years of age. As samples at the ends of the dataset's age range would be thereby involved in fewer linear models, we made the window size gradually increase from 11 to 16 years when starting from the "youngest" samples and decrease from 16 to 11 years when reaching the "oldest". The shown example to illustrate the approach is the expression of *LMO3* in subcutaneous adipose tissue over age, with 8 age windows are shown. Colored rectangles represent the age windows and colored lines the linear modelling of the gene expression over age in each of them.



Supplementary Figure 6

Clustering of pathways from Reactome and KEGG, and level 3 Gene Ontology Biological Processes.

Based on their common genes (v. Materials and Methods), pathways from curated databases were clustered in 10 families.

Supplementary Table 1:

Number of studied genes per tissue.

For each tissue, we kept the genes with at least 1 CPM of expression in at least 40% of the samples.

Tissue	#Genes
Adipose - Subcutaneous	14151
Adipose - Visceral (Omentum)	14305
Adrenal Gland	14075
Artery - Aorta	13798
Artery - Coronary	14121
Artery - Tibial	13510
Bladder	14696
Brain - Amygdala	14575
Brain - Anterior cingulate cortex (BA24)	14588
Brain - Caudate (basal ganglia)	14778
Brain - Cerebellar Hemisphere	14291
Brain - Cerebellum	14497
Brain - Cortex	14736
Brain - Frontal Cortex (BA9)	14665
Brain - Hippocampus	14627
Brain - Hypothalamus	15030
Brain - Nucleus accumbens (basal ganglia)	14756
Brain - Putamen (basal ganglia)	14516
Brain - Spinal cord (cervical c-1)	14602
Brain - Substantia nigra	14566
Breast - Mammary Tissue	14738
Cells - EBV-transformed lymphocytes	12360
Cells - Transformed fibroblasts	12787
Cervix - Ectocervix	14281
Cervix - Endocervix	15289
Colon - Sigmoid	14190
Colon - Transverse	14826
Esophagus - Gastroesophageal Junction	14030
Esophagus - Mucosa	14209
Esophagus - Muscularis	13982
Fallopian Tube	15052
Heart - Atrial Appendage	13852
Heart - Left Ventricle	13118
Kidney - Cortex	14775
Liver	13298
Lung	14953
Minor Salivary Gland	14751
Muscle - Skeletal	12473
Nerve - Tibial	14627
Ovary	14143
Pancreas	13754
Pituitary	15342
Prostate	15127
Skin - Not Sun Exposed (Suprapubic)	14635
Skin - Sun Exposed (Lower leg)	14637
Small Intestine - Terminal Ileum	15065
Spleen	14588
Stomach	14558
Testis	17965
Thyroid	14801
Uterus	14236
Vagina	14882
Whole Blood	12043

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Reviewer #1 (Public Review):

This fascinating paper by A.L. Schneider et al. describes voyAGER, a shiny-based interface for easy exploration of the GTEx dataset by non- or novice programmers. Importantly, voyAGER is open source and available from github, which could greatly accelerate additional development and further uses of this interesting tool.

The authors developed a pipeline for modeling age-related changes in gene expression in the GTEx data called ShARP-LM, fitting a linear model for age, sex, and age&sex interaction terms. This pipeline underlies the later analyses that can be applied within voyAGER. These analyses are labeled by tissue so that users can easily begin a query based on a tissue or a gene of possible interest.

voyAGER implements many kinds of interesting R-based tools such as pathway overrepresentation analysis and gene co-expression module analysis, in a way that makes these approaches accessible to non-bioinformaticist aging researchers.

As the tidal wave of publicly available large, high-dimensional datasets such as transcriptomes continues to grow exponentially, the usefulness of tools such as voyAGER will only increase. While test users may be able to imagine features or refinements they wish were already present, due to the open source approach they or anyone else including but not limited to the present authors can implement additional features in the future. I look forward to using this tool and to staying abreast of its future development.

Overall, this study describes a new tool of interest to the field. The manuscript is clearly written overall. The figures and supplementary information are all clear and all add to the manuscript.

Reviewer #2 (Public Review):

The purpose of this study is to develop a tool that serves as a starting point for investigating and uncovering genes and pathways associated with aging. The tool utilizes information from the GTEx public database, which contains post-mortem human data. It focuses on identifying age-related gene expression changes across different age range, biological sexes, and medical histories, with a focus on specific tissues.

Additionally, the authors envision the platform as continuously evolving, with ongoing development and expansion to include new data and features, ensuring it remains a cutting-edge resource for researchers studying aging.

Strengths

voyAGER presents a tool for exploring gene expression changes across multiple tissues in the context of aging. One of the main strengths of the tool is its intuitive and user-friendly interface, which allows for easy navigation and exploration of gene expression patterns for biologists. Users can explore changes in gene expression of single genes across multiple tissues, enabling them to identify genes of interest that can be further investigated.

A particularly noteworthy strength of the tool is its ability to show tissue-specific gene expression patterns. This feature is essential for elucidating the paradigm of tissue-specific asynchronous aging and provides a unique and valuable resource for the aging community.

Overall, the tool offers an entry point for further investigation of genes involved in aging, and its ability to show tissue-specific gene expression patterns provides a unique and valuable resource for the scientific community.

Lastly, the tool is accompanied by a clear and thorough tutorial that explains each of its functionalities and provides examples. The authors also acknowledge the limitations of the statistical inference tests used in the tool, which adds to its overall transparency.

Weaknesses

Underlying data analysis

In this tool/resource paper, it is crucial that the data used is up-to-date to provide the most comprehensive and relevant information to users. However, the authors utilized GTEx v7, which is an outdated (2016) version of the dataset. It is worth noting that GTEx v8 includes over 940 individuals, representing a 35% increase in individuals, and a 50% increase in the total number of samples. The authors should check the newer versions of GTEx and update the data.

The authors did not address any correction for batch effects or RNA integrity numbers, which are known to affect transcriptome profiles. For instance, our analysis of GTEx v8 Cortex tissue revealed that after filtering out lowly expressed genes, in the same way authors did, PC1 (which accounts for 24% of the variation) had a Spearman's correlation value of 0.48 ($p < 6.1 \times 10^{-16}$) with RNA integrity number.

The data analyzed in the GTEx dataset is not filtered or corrected for the cause of death, which can range from violent and sudden deaths to slow deaths or cases requiring a ventilator. As a result, the data may not accurately represent healthy aging profiles but rather reflect changes in the transcriptome specific to certain diseases due to the age-related increase in disease risk. While the authors do acknowledge this limitation in the discussion, stating that it is not a healthy cohort and disease-specific analysis is not feasible due to the limited number of samples, it would be useful for users to have the option to analyze only cases of fast death, excluding ventilator cases and deaths due to disease. This is typically how GTEx data is utilized in aging studies. Alternatively, the authors should consider including the "cause of death" variable in the model.

The age distribution varies across tissues which may impact the results of the study. The authors' claim that age distribution does not affect the outcomes is inconclusive. Since the study aims to provide cross-tissue analysis, it is important to note that differing age distributions across tissues can influence the overall results. To address this, the authors should conduct downsampling to different age distributions across tissues and evaluate the level of tissue-specific or common changes that remain after the distributions are made similar.

The GTEx resource is extremely valuable, however, it comes with challenges. GTEx contains tissue samples from the same individuals across different tissues, resulting in varying degrees of overlap in sample origin across tissues as not all tissues are collected for all individuals. This could affect the similar/different patterns observed across tissues. As this tool is meant for broader use by the community, it is crucial for the authors to either rule out this possibility by conducting a cross-tissue comparison using a non-parametric model that accounts for the dependency between samples from the same individual, or to provide information on the degree of similarity between samples so that the users can keep this possibility in mind when using the tool for hypothesis generation.

Visualisation and analysis platform

The authors aimed to create an open-source and ever-evolving resource that could be adapted and improved with new functionality. However, this goal was only partially achieved. Although the code for the web app is open source, crucial components such as the statistical tests or the linear model are not included in the repository, limiting the tool's customizability and adaptability.

Furthermore, the authors' choice of visualization platform (R shiny) may not be the best fit for extensibility and open-source collaboration, as it lacks modularity. A more suitable alternative could be production-oriented platforms such as Flask or FastAPI.

To facilitate collaboration and improve the tool's adaptability, data resulting from the pre-processing pipeline should be made publicly available. This would make it easier for others to contribute and extend the tool's functionality, ultimately enhancing its value for the scientific community.

Reviewer #3 (Public Review):

In their manuscript, Schneider et al. aim to develop voyAGER, a web-based tool that enables the exploration of gene expression changes over age in a tissue- and sex-specific manner. The authors achieved this goal by calculating the significance of gene expression alterations within a sliding window, using their unique algorithm, Shifting Age Range Pipeline for Linear Modelling (ShARP-LM), as well as tissue-level summaries that calculated the significance of the proportion of differentially expressed genes by the windows and calculated enrichments of pathways for showing biological relevance. Furthermore, the authors examined the enrichment of cell types, pathways, and diseases by defining the co-expressed gene modules in four selected tissues. The voyAGER was developed as a discovery tool, providing researchers with easy access to the vast amount of transcriptome data from the GTEx project. Overall, the research design is unique and well-performed, with interesting results that provide useful resources for the field of human genetics of aging. I have a few questions and comments, which I hope the authors can address.

1. In the gene-centric analyses section of the result, to improve this manuscript and database, linear regression tests accounting for the entire range of age should be added. The authors' algorithm, ShARP-LM, tests locally within a 16-year window which makes it has lower power than the linear regression test with the whole ages. I suspect that the power reduction is strongly affected in the younger age range since a larger number of GTEx donors are enriched in old age. By adding the results from the lm tests, readers would gain more insight and evidence into how significantly their interest genes change with age.
2. In line with the ShARP-LM test results, it is not clear which criterion was used to define the significant genes and the following enrichment analyses. I assume that the criterion is $P < 0.05$, but it should be clearly noted. Additionally, the authors should apply adjusted p-values for multiple-test correction. The ideal criterion is an adjusted $P < 0.05$. However, if

none or only a handful of genes were found to be significant, the authors could relax the criteria, such as using a regular $P < 0.01$ or 0.05 .

3. In the gene-centric analyses section, authors should provide a full list of donor conditions and a summary table of conditions as supplementary.

4. The tissue-specific assessment section has poor sub-titles. Every title has to contain information.

5. I have an issue understanding the meaning of NES from GSEA in the tissue-specific assessment section. The authors performed GSEA for the DEGs against the background genes ordered by t-statistics (from positive to negative) calculated from the linear model. I understand the p-value was two-tailed, which means that both positive and negative NES are meaningful as they represent up-regulated expression direction (positive coefficient) and down-regulated expression direction (negative coefficient) with age, respectively, within a window. However, in the GSEA section of Methods, authors were not fully elaborate on this directionality but stated, "The NES for each pathway was used in subsequent analyses as a metric of its over- or down-representation in the Peak". The authors should clearly elaborate on how to interpret the NES from their results.

6. In the Modules of co-expressed genes section, the authors did not explain how or why they selected the four tissues: brain, skeletal muscle, heart (left ventricle), and whole blood. This should be elaborated on.

7. In the modules of the co-expressed genes section, the authors did not provide an explanation of the "diseases-manual" sub-tab of the "Pathway" tab of the voyAGER tool. It would be helpful for readers to understand how the candidate disease list was prepared and what the results represent.